ELEC-237
ELECTRONICS LABORATORY-I

EXPERIMENT 3

Junction Diode Basics

Name:.....

Surname: .....

Student ID:.....

Section:.....

Date:.....

OBJECTIVE: *Familiarizing with the basic properties of junction diodes and providing an overview of some important but simple applications.*

DEPARTMENT OF ELECTRONICS ENGINEERING
GEBZE TECHNICAL UNIVERSITY

This experiment has been adopted from Department of Electrical and Electronics Engineering, Boğaziçi University

1.Diode Action

1.1 Ideal Rectification:

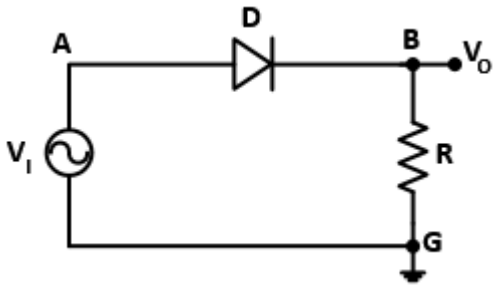


Figure 1. A circuit for the measurement of offsets.

a) Assemble the circuit as shown in Figure 1, using an IN4004 diode and 1kΩ resistor. Adjust the signal generator to provide a sine wave at 100 Hz with 20 V_{pp} amplitude. Observe and note the waveforms at nodes A and B. Indicate the voltage drop on peaks in the graph.

Waveform of .....
.....V / cm
.....ms / cm

b) Estimate the diode voltage drop v_D at the peak of the output for 20 V_{pp} and 2 V_{pp} signals.

$V_{D@10Vpeak}$ = ... $V_{D@1Vpeak}$ = ...

c) Examine the relationship between v_A and v_B near where v_B begins to go positive. Estimate the time at which the output voltage is ½ the diode drop at the peak. Note your measurements to Table 1.

Table 1. Time and phase measurement

	When v_B begins rising	When v_B is ½ diode drop
t (ms)	.....	.....

d) Switch the generator to provide a square-wave output. Notice the direct effect of the voltage drop.

Waveform of .....
.....V / cm
.....ms / cm

1.2 Rectifier Filtering:

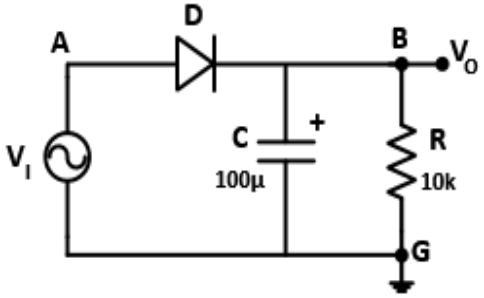


Figure 2. A Rectifier Circuit with Capacitor Filter

a) Assemble the circuit as shown in Figure 2, using an IN4004 diode. Adjust the signal generator to provide a sine wave at 100 Hz with 20 V_{pp} amplitude. Observe and note the waveforms at nodes A and B. Indicate the voltage drop on peaks in the graph. Estimate the time interval for which the diode is forward conducting (Note the values to the proper blanks below each graph).

b) Shunt R_L by a resistor R_2 = 1 kΩ. Measure v_A and v_B as in a).

Waveform of A&B for R = 10 kΩ
.....V / cm
.....ms / cm
t = ... ms

Waveform of A&B for R = (10//1) kΩ
.....V / cm
.....ms / cm
t = ... ms

c) Switch the generator to provide a square wave output. Repeat a), b) above with load R_{eq} = 10 kΩ and lkΩ // 10kΩ = 0.909 kΩ.

Waveform of A&B for R = 10 kΩ
.....V / cm
.....ms / cm
t = ... ms

Waveform of A&B for R= (10//1) kΩ
.....V / cm
.....ms / cm
t = ... ms

2.Diode Conduction – The Forward Drop

2.1 Basic Measurements:

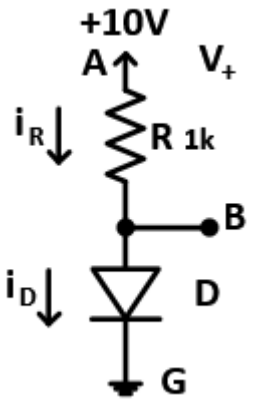


Figure 3. A Diode Forward-Drop Test Circuit

Assemble the circuit shown in Figure 3, using a IN4004 diode, a l kΩ resistor

and a 10 V de supply. Note that the circuit is essentially the same as that in Figure 1, but with a dc signal source, and the location of circuit ground redefined.

a) Measure v_B. Adjust the supply to 10.0(0) V (for convenience). Find i_D.

b) Shunt *R* with a resistor of equal value 1 kΩ. Measure $v_B = v_D$. Find i_D.

c) With two 1 kΩ resistors connected, shunt D with a second IN4004 diode (assumed to be matched). What does v_D become?

Table 2. Voltage Measurements and Current Estimates

	R = 1 kΩ Single Diode	R = (1 // 1) kΩ Single Diode	R = (1 // 1) kΩ Two Diodes Shunted
v_B (= v_D)	.....	.....	.....
i_D (calculated)	.....	.....	.....

2.2 Forward Conduction Modelling – Finding a Large Signal Model:

a) Return to the circuit in Figure 3, but use four wide-ranging values for R, namely 1 kΩ, 10 kΩ, 100 kΩ, and l MΩ and a IN4004 diode. Experimentally, ground the cathode of the diode to be tested, and connect all four resistors to its anode with one end of each open, and to which the 10 V supply will be connected in turn. For a supply voltage of 10 V, measure V_D as each resistor is connected to the supply in turn.

Approximate the corresponding current. Note that if the supply is raised slightly, to 10.7 or so, the values of current become somewhat "friendlier", being easier to estimate (or calculate)! *(Hint: Observe the exponential relation between V_D and I_D.)*

b) Repeat for a second diode type a IN914.

Table 3. Voltage Measurements and Current Estimates for IN4004 (Large Signal Model)

IN4004		R = 1 kΩ	R = 10 kΩ	R = 100 kΩ	R = 1 MΩ
V_{DD} = 10 V	V_D (meas.)	.....	.....	.....	.....
	I_D (calc.)	.....	.....	.....	.....
V_{DD} = 10.7 V	V_D (meas.)	.....	.....	.....	.....
	I_D (calc.)	.....	.....	.....	.....

Table 4. Voltage Measurements and Current Estimates for IN914 (Large Signal Model)

IN914		R = 1 kΩ	R = 10 kΩ	R = 100 kΩ	R = 1 MΩ
V_{DD} = 10 V	V_D (meas.)	.....	.....	.....	.....
	I_D (calc.)	.....	.....	.....	.....
V_{DD} = 10.7 V	V_D (meas.)	.....	.....	.....	.....
	I_D (calc.)	.....	.....	.....	.....

2.3 Forward Conduction Modelling – Finding a Small Signal Model:

a) Connect a IN914 diode using V^+ =10 V and R =1 kΩ. Measure V_D.
b) Shunt the diode by a resistor which reduces the diode current by 5% to 10%. Notice, in general, that an appropriate value is R. In particular, here use l kΩ. Measure V_D. Calculate an estimate for r_d at nearly 10 mA.
c) Repeat the procedure above for R =100 kΩ, and find r_d at nearly 100 μA.
d) Repeat all of the above for a IN4004 diode.

Table 6. Voltage Measurements and r_D Estimates for IN914 and IN4004 (Small Signal Model)

	R = 1 kΩ			R = 100 kΩ		
	V_D with no R shunting	V_D with R shunting	r_D @ ~10 mA	V_D with no R shunting	V_D with R shunting	r_D @ ~0.1 mA
IN4004	.....	.....	.....	.....	.....	.....
IN914	.....	.....	.....	.....	.....	.....

NOTES